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الجامعة الافتراضية السورية

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وظيفة الخوارزميات الذكية

BIA601

أعدّ هذا التقرير بتكليف من الدكتور:

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Optimizing Recommendations in E-commerce Stores using Genetic Algorithms

1. Introduction and Scientific Reference

This project is grounded in contemporary research on evolutionary intelligence. The foundational methodology is derived from:

> Deshmukh, et al. (2024). *"Customer behavioural content recommendation system using decision tree and genetic algorithm for online shopping websites"*. *Multidisciplinary Science Journal*.

The implementation shifts from static filtering to a dynamic, evolutionary approach where recommendation sets are treated as "Chromosomes" that evolve based on real-time user DNA (Age, Location) and interaction history.

2. Technology Stack & Dataset

- Dataset: Built and tested using the official HW__Data_S25 university dataset.
- Frontend: React (Vite) with a custom "Evolutionary UI" for real-time population updates.
- Backend: Python (FastAPI) for high-speed genetic computations.
- Data Engine: Pandas for merging four distinct datasets: `users.xlsx`, `products.xlsx`, `ratings.xlsx`, and `behavior_15500.xlsx`.

3. The Genetic Recommendation Logic

Unlike traditional systems, this AI simulates Natural Selection to manage store resources:

A. The Biological Mapping

- **Population**: Multiple sets of recommendation lists (Set Alpha, Beta, Gamma).
- **Chromosome**: A single list of 5 products tailored to a specific `user_id`.
- **Fitness Function**: The core "Survival" metric calculated as:
  $$Fitness = (Purchased \times 10) + (Clicked \times 5) + (Viewed \times 1) + Avg_Rating$$

B. Evolutionary Phases in our Implementation

1. **Selection**: The system identifies products with the highest fitness for the user's specific age group and location.
2. **Crossover (Hybridization)**: Mixing high-performing products from different categories to create a diverse "Offspring" list.
3. **Mutation (The Discovery Engine)**: When a user rejects a product (clicks "Not Interested"), the system triggers a **Mutation**. It injects a new, high-fitness gene (product) that the user has never seen before, preventing "Local Optimum" (the echo-chamber effect).

4. Key Prototype Features

- **User DNA Awareness**: Recommendations dynamically shift when switching between different user profiles (e.g., a 20-year-old in New York vs. a 50-year-old in Qatar).
- **Social Proofing**: Real-time badges like Trending in [City] generated by cross-referencing `behavior` and `users` tables.
- **Visual Evolution**: A Evolving Generation state that visually demonstrates the mutation process to the end-user.